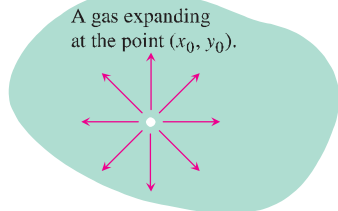


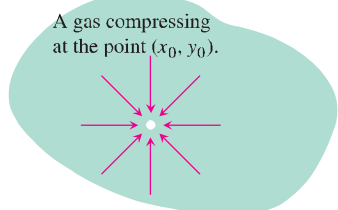
**Source:**  $\operatorname{div} \mathbf{F}(x_0, y_0) > 0$

A gas expanding  
at the point  $(x_0, y_0)$ .



**Sink:**  $\operatorname{div} \mathbf{F}(x_0, y_0) < 0$

A gas compressing  
at the point  $(x_0, y_0)$ .



**FIGURE 16.27** If a gas is expanding at a point  $(x_0, y_0)$ , the lines of flow have positive divergence; if the gas is compressing, the divergence is negative.

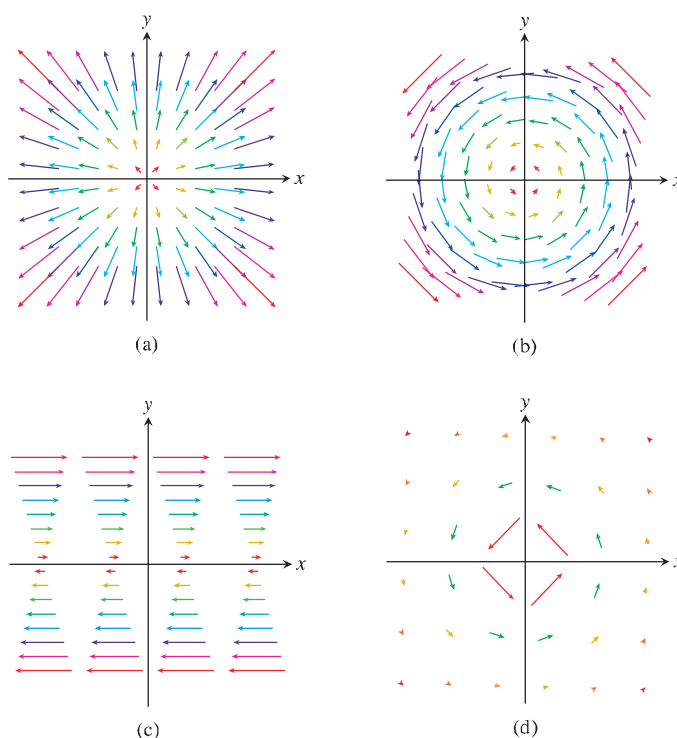
Finally, we let  $\Delta x$  and  $\Delta y$  approach zero to define the flux density of  $\mathbf{F}$  at the point  $(x, y)$ . In mathematics, we call the flux density the *divergence* of  $\mathbf{F}$ . The symbol for it is  $\operatorname{div} \mathbf{F}$ , pronounced “divergence of  $\mathbf{F}$ ” or “div  $\mathbf{F}$ .”

**DEFINITION** The **divergence (flux density)** of a vector field  $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$  at the point  $(x, y)$  is

$$\operatorname{div} \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}. \quad (1)$$

A gas is compressible, unlike a liquid, and the divergence of its velocity field measures to what extent it is expanding or compressing at each point. Intuitively, if a gas is expanding at the point  $(x_0, y_0)$ , the lines of flow would diverge there (hence the name) and, since the gas would be flowing out of a small rectangle about  $(x_0, y_0)$ , the divergence of  $\mathbf{F}$  at  $(x_0, y_0)$  would be positive. If the gas were compressing instead of expanding, the divergence would be negative (Figure 16.27).

**EXAMPLE 1** The following vector fields represent the velocity of a gas flowing in the  $xy$ -plane. Find the divergence of each vector field and interpret its physical meaning. Figure 16.28 displays the vector fields.



**FIGURE 16.28** Velocity fields of a gas flowing in the plane (Example 1).

(a) *Uniform expansion or compression:*  $\mathbf{F}(x, y) = cx\mathbf{i} + cy\mathbf{j}$

(b) *Uniform rotation:*  $\mathbf{F}(x, y) = -cy\mathbf{i} + cx\mathbf{j}$

(c) *Shearing flow:*  $\mathbf{F}(x, y) = y\mathbf{i}$

(d) *Whirlpool effect:*  $\mathbf{F}(x, y) = \frac{-y}{x^2 + y^2} \mathbf{i} + \frac{x}{x^2 + y^2} \mathbf{j}$

**Solution**

- (a)  $\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(cx) + \frac{\partial}{\partial y}(cy) = 2c$ : If  $c > 0$ , the gas is undergoing uniform expansion; if  $c < 0$ , it is undergoing uniform compression.
- (b)  $\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(-cy) + \frac{\partial}{\partial y}(cx) = 0$ : The gas is neither expanding nor compressing.
- (c)  $\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(y) = 0$ : The gas is neither expanding nor compressing.
- (d)  $\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}\left(\frac{-y}{x^2 + y^2}\right) + \frac{\partial}{\partial y}\left(\frac{x}{x^2 + y^2}\right) = \frac{2xy}{(x^2 + y^2)^2} - \frac{2xy}{(x^2 + y^2)^2} = 0$ : Again, the divergence is zero at all points in the domain of the velocity field.

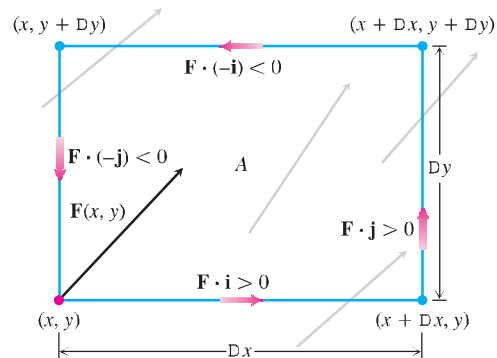
Cases (b), (c), and (d) of Figure 16.28 are plausible models for the two-dimensional flow of a liquid. In fluid dynamics, when the velocity field of a flowing liquid always has divergence equal to zero, as in those cases, the liquid is said to be **incompressible**.

**Spin Around an Axis: The k-Component of Curl**

The second idea we need for Green's Theorem has to do with measuring how a floating paddle wheel, with axis perpendicular to the plane, spins at a point in a fluid flowing in a plane region. This idea gives some sense of how the fluid is circulating around axes located at different points and perpendicular to the region. Physicists sometimes refer to this as the *circulation density* of a vector field  $\mathbf{F}$  at a point. To obtain it, we return to the velocity field

$$\mathbf{F}(x, y) = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$$

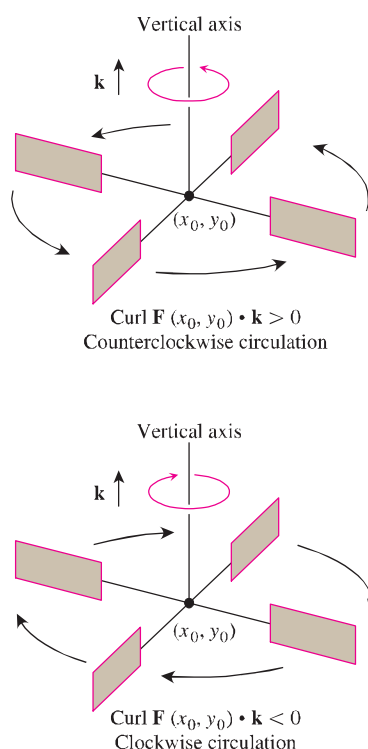
and consider the rectangle  $A$  in Figure 16.29 (where we assume both components of  $\mathbf{F}$  are positive).



**FIGURE 16.29** The rate at which a fluid flows along the bottom edge of a rectangular region  $A$  in the direction  $\mathbf{i}$  is approximately  $\mathbf{F}(x, y) \cdot \mathbf{i} \Delta x$ , which is positive for the vector field  $\mathbf{F}$  shown here. To approximate the rate of circulation at the point  $(x, y)$ , we calculate the (approximate) flow rates along each edge in the directions of the red arrows, sum these rates, and then divide the sum by the area of  $A$ . Taking the limit as  $\Delta x \rightarrow 0$  and  $\Delta y \rightarrow 0$  gives the rate of the circulation per unit area.

The circulation rate of  $\mathbf{F}$  around the boundary of  $A$  is the sum of flow rates along the sides in the tangential direction. For the bottom edge, the flow rate is approximately

$$\mathbf{F}(x, y) \cdot \mathbf{i} \Delta x = M(x, y) \Delta x.$$



**FIGURE 16.30** In the flow of an incompressible fluid over a plane region, the  $\mathbf{k}$ -component of the curl measures the rate of the fluid's rotation at a point. The  $\mathbf{k}$ -component of the curl is positive at points where the rotation is counterclockwise and negative where the rotation is clockwise.

This is the scalar component of the velocity  $\mathbf{F}(x, y)$  in the tangent direction  $\mathbf{i}$  times the length of the segment. The flow rates may be positive or negative depending on the components of  $\mathbf{F}$ . We approximate the net circulation rate around the rectangular boundary of  $A$  by summing the flow rates along the four edges as defined by the following dot products.

$$\begin{aligned} \text{Top:} \quad & \mathbf{F}(x, y + \Delta y) \cdot (-\mathbf{i}) \Delta x = -M(x, y + \Delta y) \Delta x \\ \text{Bottom:} \quad & \mathbf{F}(x, y) \cdot \mathbf{i} \Delta x = M(x, y) \Delta x \\ \text{Right:} \quad & \mathbf{F}(x + \Delta x, y) \cdot \mathbf{j} \Delta y = N(x + \Delta x, y) \Delta y \\ \text{Left:} \quad & \mathbf{F}(x, y) \cdot (-\mathbf{j}) \Delta y = -N(x, y) \Delta y. \end{aligned}$$

We sum opposite pairs to get

$$\text{Top and bottom:} \quad -(M(x, y + \Delta y) - M(x, y)) \Delta x \approx -\left(\frac{\partial M}{\partial y} \Delta y\right) \Delta x$$

$$\text{Right and left:} \quad (N(x + \Delta x, y) - N(x, y)) \Delta y \approx \left(\frac{\partial N}{\partial x} \Delta x\right) \Delta y.$$

Adding these last two equations gives the net circulation relative to the counterclockwise orientation, and dividing by  $\Delta x \Delta y$  gives an estimate of the circulation density for the rectangle:

$$\frac{\text{Circulation around rectangle}}{\text{rectangle area}} \approx \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}.$$

We let  $\Delta x$  and  $\Delta y$  approach zero to define the *circulation density* of  $\mathbf{F}$  at the point  $(x, y)$ .

If we see a counterclockwise rotation looking downward onto the  $xy$ -plane from the tip of the unit  $\mathbf{k}$  vector, then the circulation density is positive (Figure 16.30). The value of the circulation density is the  $\mathbf{k}$ -component of a more general circulation vector field we define in Section 16.7, called the *curl* of the vector field  $\mathbf{F}$ . For Green's Theorem, we need only this  $\mathbf{k}$ -component.

**DEFINITION** The **circulation density** of a vector field  $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$  at the point  $(x, y)$  is the scalar expression

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}. \quad (2)$$

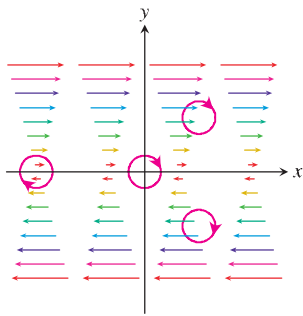
This expression is also called **the  $\mathbf{k}$ -component of the curl**, denoted by  $(\text{curl } \mathbf{F}) \cdot \mathbf{k}$ .

If water is moving about a region in the  $xy$ -plane in a thin layer, then the  $\mathbf{k}$ -component of the curl at a point  $(x_0, y_0)$  gives a way to measure how fast and in what direction a small paddle wheel spins if it is put into the water at  $(x_0, y_0)$  with its axis perpendicular to the plane, parallel to  $\mathbf{k}$  (Figure 16.30).

**EXAMPLE 2** Find the circulation density, and interpret what it means, for each vector field in Example 1.

**Solution**

- (a) *Uniform expansion:*  $(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \frac{\partial}{\partial x}(cy) - \frac{\partial}{\partial y}(cx) = 0$ . The gas is not circulating at very small scales.



**FIGURE 16.31** A shearing flow pushes the fluid clockwise around each point (Example 2c).

- (b) *Rotation:*  $(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \frac{\partial}{\partial x}(cx) - \frac{\partial}{\partial y}(-cy) = 2c$ . The constant circulation density indicates rotation at every point. If  $c > 0$ , the rotation is counterclockwise; if  $c < 0$ , the rotation is clockwise.
- (c) *Shear:*  $(\text{curl } \mathbf{F}) \cdot \mathbf{k} = -\frac{\partial}{\partial y}(y) = -1$ . The circulation density is constant and negative, so a paddle wheel floating in water undergoing such a shearing flow spins clockwise. The rate of rotation is the same at each point. The average effect of the fluid flow is to push fluid clockwise around each of the small circles shown in Figure 16.31.
- (d) *Whirlpool:*

$$(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \frac{\partial}{\partial x} \left( \frac{x}{x^2 + y^2} \right) - \frac{\partial}{\partial y} \left( \frac{-y}{x^2 + y^2} \right) = \frac{y^2 - x^2}{(x^2 + y^2)^2} - \frac{y^2 - x^2}{(x^2 + y^2)^2} = 0.$$

The circulation density is 0 at every point away from the origin (where the vector field is undefined and the whirlpool effect is taking place), and the gas is not circulating at any point for which the vector field is defined. ■

## Two Forms for Green's Theorem

In one form, Green's Theorem says that under suitable conditions the outward flux of a vector field across a simple closed curve in the plane equals the double integral of the divergence of the field over the region enclosed by the curve. Recall the formulas for flux in Equations (3) and (4) in Section 16.2 and that a curve is simple if it does not cross itself.

**THEOREM 4—Green's Theorem (Flux-Divergence or Normal Form)** Let  $C$  be a piecewise smooth, simple closed curve enclosing a region  $R$  in the plane. Let  $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$  be a vector field with  $M$  and  $N$  having continuous first partial derivatives in an open region containing  $R$ . Then the outward flux of  $\mathbf{F}$  across  $C$  equals the double integral of  $\text{div } \mathbf{F}$  over the region  $R$  enclosed by  $C$ .

$$\oint_C \mathbf{F} \cdot \mathbf{n} \, ds = \oint_C M \, dy - N \, dx = \iint_R \left( \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx \, dy \quad (3)$$

Outward flux Divergence integral

We introduced the notation  $\oint_C$  in Section 16.3 for integration around a closed curve. We elaborate further on the notation here. A simple closed curve  $C$  can be traversed in two possible directions. The curve is traversed counterclockwise, and said to be *positively oriented*, if the region it encloses is always to the left of an object as it moves along the path. Otherwise it is traversed clockwise and *negatively oriented*. The line integral of a vector field  $\mathbf{F}$  along  $C$  reverses sign if we change the orientation. We use the notation

$$\oint_C \mathbf{F}(x, y) \cdot d\mathbf{r}$$

for the line integral when the simple closed curve  $C$  is traversed counterclockwise, with its positive orientation.

A second form of Green's Theorem says that the counterclockwise circulation of a vector field around a simple closed curve is the double integral of the  $\mathbf{k}$ -component of the curl of the field over the region enclosed by the curve. Recall the defining Equation (2) for circulation in Section 16.2.

**THEOREM 5—Green's Theorem (Circulation-Curl or Tangential Form)**

Let  $C$  be a piecewise smooth, simple closed curve enclosing a region  $R$  in the plane. Let  $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$  be a vector field with  $M$  and  $N$  having continuous first partial derivatives in an open region containing  $R$ . Then the counterclockwise circulation of  $\mathbf{F}$  around  $C$  equals the double integral of  $(\text{curl } \mathbf{F}) \cdot \mathbf{k}$  over  $R$ .

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \oint_C M \, dx + N \, dy = \iint_R \left( \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx \, dy \quad (4)$$

Counterclockwise circulation

Curl integral

The two forms of Green's Theorem are equivalent. Applying Equation (3) to the field  $\mathbf{G}_1 = N\mathbf{i} - M\mathbf{j}$  gives Equation (4), and applying Equation (4) to  $\mathbf{G}_2 = -N\mathbf{i} + M\mathbf{j}$  gives Equation (3).

Both forms of Green's Theorem can be viewed as two-dimensional generalizations of the Net Change Theorem in Section 5.4. The outward flux of  $\mathbf{F}$  across  $C$ , defined by the line integral on the left-hand side of Equation (3), is the integral of its rate of change (flux density) over the region  $R$  enclosed by  $C$ , which is the double integral on the right-hand side of Equation (3). Likewise, the counterclockwise circulation of  $\mathbf{F}$  around  $C$ , defined by the line integral on the left-hand side of Equation (4), is the integral of its rate of change (circulation density) over the region  $R$  enclosed by  $C$ , which is the double integral on the right-hand side of Equation (4).

**EXAMPLE 3** Verify both forms of Green's Theorem for the vector field

$$\mathbf{F}(x, y) = (x - y)\mathbf{i} + x\mathbf{j}$$

and the region  $R$  bounded by the unit circle

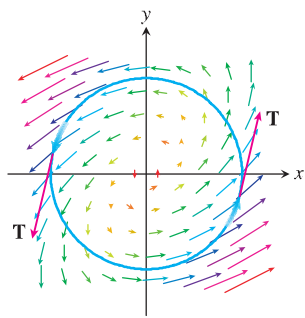
$$C: \mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j}, \quad 0 \leq t \leq 2\pi.$$

**Solution** Evaluating  $\mathbf{F}(\mathbf{r}(t))$  and differentiating components, we have

$$\begin{aligned} M &= \cos t - \sin t, & dx &= d(\cos t) = -\sin t \, dt, \\ N &= \cos t, & dy &= d(\sin t) = \cos t \, dt, \\ \frac{\partial M}{\partial x} &= 1, & \frac{\partial M}{\partial y} &= -1, & \frac{\partial N}{\partial x} &= 1, & \frac{\partial N}{\partial y} &= 0. \end{aligned}$$

The two sides of Equation (3) are

$$\begin{aligned} \oint_C M \, dy - N \, dx &= \int_{t=0}^{t=2\pi} (\cos t - \sin t)(\cos t \, dt) - (\cos t)(-\sin t \, dt) \\ &= \int_0^{2\pi} \cos^2 t \, dt = \pi \\ \iint_R \left( \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx \, dy &= \iint_R (1 + 0) \, dx \, dy \\ &= \iint_R dx \, dy = \text{area inside the unit circle} = \pi. \end{aligned}$$



**FIGURE 16.32** The vector field in Example 3 has a counterclockwise circulation of  $2\pi$  around the unit circle.

The two sides of Equation (4) are

$$\begin{aligned}\oint_C M dx + N dy &= \int_{t=0}^{t=2\pi} (\cos t - \sin t)(-\sin t dt) + (\cos t)(\cos t dt) \\ &= \int_0^{2\pi} (-\sin t \cos t + 1) dt = 2\pi \\ \iint_R \left( \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \iint_R (1 - (-1)) dx dy = 2 \iint_R dx dy = 2\pi.\end{aligned}$$

Figure 16.32 displays the vector field and circulation around  $C$ . ■

### Using Green's Theorem to Evaluate Line Integrals

If we construct a closed curve  $C$  by piecing together a number of different curves end to end, the process of evaluating a line integral over  $C$  can be lengthy because there are so many different integrals to evaluate. If  $C$  bounds a region  $R$  to which Green's Theorem applies, however, we can use Green's Theorem to change the line integral around  $C$  into one double integral over  $R$ .

#### EXAMPLE 4 Evaluate the line integral

$$\oint_C xy dy - y^2 dx,$$

where  $C$  is the square cut from the first quadrant by the lines  $x = 1$  and  $y = 1$ .

**Solution** We can use either form of Green's Theorem to change the line integral into a double integral over the square.

1. *With the Normal Form* Equation (3): Taking  $M = xy$ ,  $N = y^2$ , and  $C$  and  $R$  as the square's boundary and interior gives

$$\begin{aligned}\oint_C xy dy - y^2 dx &= \iint_R (y + 2y) dx dy = \int_0^1 \int_0^1 3y dx dy \\ &= \int_0^1 \left[ 3xy \right]_{x=0}^{x=1} dy = \int_0^1 3y dy = \left[ \frac{3}{2} y^2 \right]_0^1 = \frac{3}{2}.\end{aligned}$$

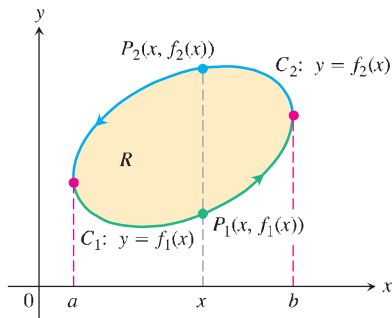
2. *With the Tangential Form* Equation (4): Taking  $M = -y^2$  and  $N = xy$  gives the same result:

$$\oint_C -y^2 dx + xy dy = \iint_R (y - (-2y)) dx dy = \frac{3}{2}.$$
 ■

**EXAMPLE 5** Calculate the outward flux of the vector field  $\mathbf{F}(x, y) = x\mathbf{i} + y^2\mathbf{j}$  across the square bounded by the lines  $x = \pm 1$  and  $y = \pm 1$ .

**Solution** Calculating the flux with a line integral would take four integrations, one for each side of the square. With Green's Theorem, we can change the line integral to one double integral. With  $M = x$ ,  $N = y^2$ ,  $C$  the square, and  $R$  the square's interior, we have

$$\begin{aligned}\text{Flux} &= \oint_C \mathbf{F} \cdot \mathbf{n} \, ds = \oint_C M \, dy - N \, dx \\ &= \iint_R \left( \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx \, dy \quad \text{Green's Theorem} \\ &= \int_{-1}^1 \int_{-1}^1 (1 + 2y) \, dx \, dy = \int_{-1}^1 \left[ x + 2xy \right]_{x=-1}^{x=1} dy \\ &= \int_{-1}^1 (2 + 4y) \, dy = \left[ 2y + 2y^2 \right]_{-1}^1 = 4. \quad \blacksquare\end{aligned}$$



**FIGURE 16.33** The boundary curve  $C$  is made up of  $C_1$ , the graph of  $y = f_1(x)$ , and  $C_2$ , the graph of  $y = f_2(x)$ .

### Proof of Green's Theorem for Special Regions

Let  $C$  be a smooth simple closed curve in the  $xy$ -plane with the property that lines parallel to the axes cut it at no more than two points. Let  $R$  be the region enclosed by  $C$  and suppose that  $M$ ,  $N$ , and their first partial derivatives are continuous at every point of some open region containing  $C$  and  $R$ . We want to prove the circulation-curl form of Green's Theorem,

$$\oint_C M \, dx + N \, dy = \iint_R \left( \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx \, dy. \quad (5)$$

Figure 16.33 shows  $C$  made up of two directed parts:

$$C_1: y = f_1(x), \quad a \leq x \leq b, \quad C_2: y = f_2(x), \quad b \geq x \geq a.$$

For any  $x$  between  $a$  and  $b$ , we can integrate  $\partial M / \partial y$  with respect to  $y$  from  $y = f_1(x)$  to  $y = f_2(x)$  and obtain

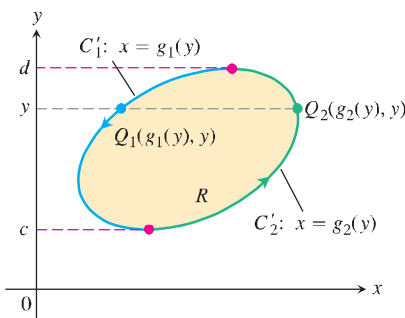
$$\int_{f_1(x)}^{f_2(x)} \frac{\partial M}{\partial y} \, dy = M(x, y) \Big|_{y=f_1(x)}^{y=f_2(x)} = M(x, f_2(x)) - M(x, f_1(x)).$$

We can then integrate this with respect to  $x$  from  $a$  to  $b$ :

$$\begin{aligned}\int_a^b \int_{f_1(x)}^{f_2(x)} \frac{\partial M}{\partial y} \, dy \, dx &= \int_a^b [M(x, f_2(x)) - M(x, f_1(x))] \, dx \\ &= - \int_b^a M(x, f_2(x)) \, dx - \int_a^b M(x, f_1(x)) \, dx \\ &= - \int_{C_2} M \, dx - \int_{C_1} M \, dx \\ &= - \oint_C M \, dx.\end{aligned}$$

Therefore

$$\oint_C M \, dx = \iint_R \left( -\frac{\partial M}{\partial y} \right) dx \, dy. \quad (6)$$



**FIGURE 16.34** The boundary curve  $C$  is made up of  $C_1'$ , the graph of  $x = g_1(y)$ , and  $C_2'$ , the graph of  $x = g_2(y)$ .

Equation (6) is half the result we need for Equation (5). We derive the other half by integrating  $\partial N / \partial x$  first with respect to  $x$  and then with respect to  $y$ , as suggested by Figure 16.34.

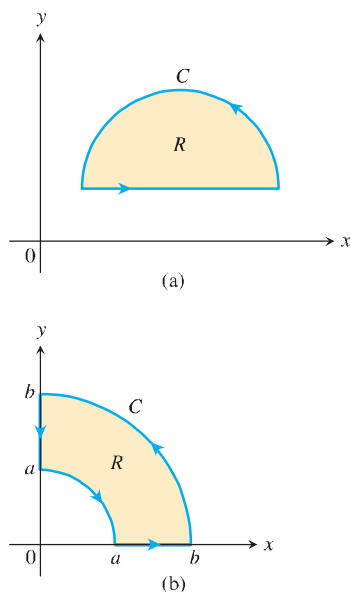


FIGURE 16.35 Other regions to which Green's Theorem applies.

This shows the curve  $C$  of Figure 16.33 decomposed into the two directed parts  $C'_1: x = g_1(y), d \geq y \geq c$  and  $C'_2: x = g_2(y), c \leq y \leq d$ . The result of this double integration is

$$\oint_C N dy = \iint_R \frac{\partial N}{\partial x} dx dy. \quad (7)$$

Summing Equations (6) and (7) gives Equation (5). This concludes the proof. ■

Green's Theorem also holds for more general regions, such as those shown in Figures 16.35 and 16.36, but we will not prove this result here. Notice that the region in Figure 16.36 is not simply connected. The curves  $C_1$  and  $C_h$  on its boundary are oriented so that the region  $R$  is always on the left-hand side as the curves are traversed in the directions shown. With this convention, Green's Theorem is valid for regions that are not simply connected.

While we stated the theorem in the  $xy$ -plane, Green's Theorem applies to any region  $R$  contained in a plane bounded by a curve  $C$  in space. We will see how to express the double integral over  $R$  for this more general form of Green's Theorem in Section 16.7.

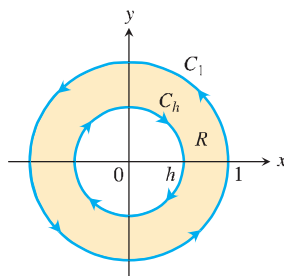


FIGURE 16.36 Green's Theorem may be applied to the annular region  $R$  by summing the line integrals along the boundaries  $C_1$  and  $C_h$  in the directions shown.

## Exercises 16.4

### Verifying Green's Theorem

In Exercises 1–4, verify the conclusion of Green's Theorem by evaluating both sides of Equations (3) and (4) for the field  $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ . Take the domains of integration in each case to be the disk  $R: x^2 + y^2 \leq a^2$  and its bounding circle  $C: \mathbf{r} = (a \cos t)\mathbf{i} + (a \sin t)\mathbf{j}, 0 \leq t \leq 2\pi$ .

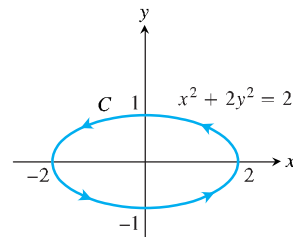
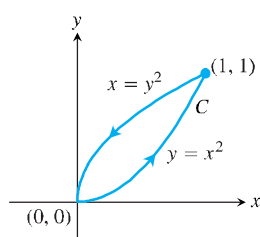
1.  $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$
2.  $\mathbf{F} = y\mathbf{i}$
3.  $\mathbf{F} = 2x\mathbf{i} - 3y\mathbf{j}$
4.  $\mathbf{F} = -x^2y\mathbf{i} + xy^2\mathbf{j}$

### Circulation and Flux

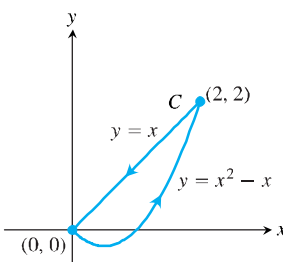
In Exercises 5–14, use Green's Theorem to find the counterclockwise circulation and outward flux for the field  $\mathbf{F}$  and curve  $C$ .

5.  $\mathbf{F} = (x - y)\mathbf{i} + (y - x)\mathbf{j}$   
 $C$ : The square bounded by  $x = 0, x = 1, y = 0, y = 1$
6.  $\mathbf{F} = (x^2 + 4y)\mathbf{i} + (x + y^2)\mathbf{j}$   
 $C$ : The square bounded by  $x = 0, x = 1, y = 0, y = 1$
7.  $\mathbf{F} = (y^2 - x^2)\mathbf{i} + (x^2 + y^2)\mathbf{j}$   
 $C$ : The triangle bounded by  $y = 0, x = 3$ , and  $y = x$
8.  $\mathbf{F} = (x + y)\mathbf{i} - (x^2 + y^2)\mathbf{j}$   
 $C$ : The triangle bounded by  $y = 0, x = 1$ , and  $y = x$

9.  $\mathbf{F} = (xy + y^2)\mathbf{i} + (x - y)\mathbf{j}$
10.  $\mathbf{F} = (x + 3y)\mathbf{i} + (2x - y)\mathbf{j}$



11.  $\mathbf{F} = x^3y^2\mathbf{i} + \frac{1}{2}x^4y\mathbf{j}$



12.  $\mathbf{F} = \frac{x}{1 + y^2}\mathbf{i} + (\tan^{-1} y)\mathbf{j}$

